Exercise J-M

1. Let $p, q \in R$ and $\left(1-\sqrt{3}i\right)^{200} = 2^{199}\left(p+iq\right)$, $i = \sqrt{-1}$. Then $p + q + q^2$ and $p - q + q^2$ are roots of the equation : **[JEE(Main)Jan.2023]**

माना $p, q \in R$ तथा $\left(1-\sqrt{3}i\right)^{200} = 2^{199}\left(p+iq\right)$, $i = \sqrt{-1}$ तो $p + q + q^2$ तथा $p - q + q^2$ किस समीकरण के मूल है : **[JEE(Main)Jan.2023]**

(A) $x^2 - 4x - 1 = 0$ (B) $x^2 - 4x + 1 = 0$ (C) $x^2 + 4x + 1 = 0$ (D) $x^2 + 4x - 1 = 0$

Ans. B

Sol. $(1+\sqrt{3}i)^{200} = 2^{199}(p+iq)$

$$\Rightarrow 2^{200}\left(\frac{1}{2}+\frac{\sqrt{3}}{2}i\right)^{200} = 2^{199}(p+iq)$$

$$\Rightarrow 2^{200}\left(\cos\left(\frac{\pi}{3}\right)+i\sin\left(\frac{\pi}{3}\right)\right)^{200} = 2^{199}(p+iq)$$

$$\Rightarrow 2^{200}\left(e^{i\frac{\pi}{2}}\right)^{200} = 2^{199}(p+iq)$$

$$\Rightarrow 2^{200}\left(\cos\left(\frac{200\pi}{3}\right)+i\sin\left(\frac{200\pi}{3}\right)\right) = 2^{199}(p+iq)$$

$$\Rightarrow 2^{200}\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}i\right) = 2^{199}(p+iq)$$

$$\Rightarrow 2^{199}(-1+\sqrt{3}i) = 2^{199}(p+iq)$$

$$\Rightarrow p = -1, q = \sqrt{3}$$

$$p + q + q^2 = 2 + \sqrt{3}$$

$$p - q + q^2 = 2 - \sqrt{3}$$

2. The value of $\left(\frac{1+\sin\frac{2\pi}{9}+i\cos\frac{2\pi}{9}}{1+\sin\frac{2\pi}{9}-i\cos\frac{2\pi}{9}}\right)^3$ is : **[JEE(Main)Jan.2023]**

$\left(\frac{1+\sin\frac{2\pi}{9}+i\cos\frac{2\pi}{9}}{1+\sin\frac{2\pi}{9}-i\cos\frac{2\pi}{9}}\right)^3$ का मान है :

(A) $-\frac{1}{2}\left(\sqrt{3}-i\right)$ (B) $-\frac{1}{2}\left(1-i\sqrt{3}\right)$ (C) $\frac{1}{2}\left(1-i\sqrt{3}\right)$ (D) $\frac{1}{2}\left(\sqrt{3}+i\right)$

Ans. A

Sol. Let $\sin\frac{2\pi}{9}+i\cos\frac{2\pi}{9}=z$

$$\left(\frac{1+z}{1+\bar{z}}\right)^3=\left(\frac{1+z}{1+\frac{1}{z}}\right)^3=z^3$$

$$\Rightarrow\left(i\left(\cos\frac{2\pi}{9}-i\sin\frac{2\pi}{9}\right)\right)^3$$

$$=-i\left(\cos\frac{2\pi}{3}-i\sin\frac{2\pi}{3}\right)=-i\left(\frac{-1}{2}-i\frac{\sqrt{3}}{2}\right)$$

$$\Rightarrow\frac{-1}{2}(\sqrt{3}-i)$$

3. Let $z_1=2+3i$ and $z_2=3+4i$. The set $S=\left\{z\in C:|z-z_1|^2-|z-z_2|^2=|z_1-z_2|^2\right\}$ represents a :

[JEE(Main)Jan.2023]

(A) straight line with the sum of its intercepts on the coordinate axes equals –18

(B) hyperbola with eccentricity 2

(C) straight line with the sum of its intercepts on the coordinate axes equals 14

(D) hyperbola with the length of the transverse axis 7

माना $z_1=2+3i$ तथा $z_2=3+4i$ है। तो समुच्चय $S=\left\{z\in C:|z-z_1|^2-|z-z_2|^2=|z_1-z_2|^2\right\}$ किस को निरूपित करता है :

[JEE(Main)Jan.2023]

(A) एक सरल रेखा जिसके निर्देशांक अक्षों पर अंतःखंडों का योग –18 है

(B) एक अतिपरवलय जिसकी उत्केन्द्रता 2 है

(C) एक सरल रेखा जिसके निर्देशांक अक्षों पर अंतःखंडों का योग 14 है

(D) अतिपरवलय जिसके अनुप्रस्थ अक्ष की लंबाई 7 है

Ans. C

Sol. $\left((x-2)^2+(y-3)^2\right)-\left((x-3)^2-(y-4)^2\right)=1+1$

$\Rightarrow x+y=7$

4. Let z be a complex number such that $\left|\frac{z-2i}{z+i}\right|=2, z \neq -i$. Then z lies on the circle of radius 2 and centre :

[JEE(Main)Jan.2023]

माना z एक सम्मिश्र संख्या है तथा $\left|\frac{z-2i}{z+i}\right|=2, z \neq -i$ है। तो z त्रिज्या 2 के एक वृत्त पर है जिसका केन्द्र है

[JEE(Main)Jan.2023]

(A) (2,0) (B) (0, –2) (C) (0,2) (D) (0,0)

Ans. B

Sol. $(z-2i)(\overline{z}+2i)=4(z+i)(\overline{z}-i)$

$z\overline{z}+4+2i(z-\overline{z})=4(z\overline{z}+1+i(\overline{z}-z))$

$3z\overline{z}-6i(z-\overline{z})=0$

$x^2+y^2-2i(2iy)=0$

$x^2+y^2+4y=0$

5. For two non-zero complex numbers z_1 and z_2, if $\text{Re}(z_1z_2)=0$ and $\text{Re}(z_1+z_2)=0$, then which of the following are possible?

A. $\text{Im}(z_1)>0$ and $\text{Im}(z_2)>0$

B. $\text{Im}(z_1)<0$ and $\text{Im}(z_2)>0$

C. $\text{Im}(z_1)>0$ and $\text{Im}(z_2)<0$

D. $\text{Im}(z_1)<0$ and $\text{Im}(z_2)<0$

Choose the correct answer from the options given below : **[JEE(Main)Jan.2023]**

(A) B and C (B) B and D (C) A and B (D) A and C

z_1 व z_2 दो शून्येतर सम्मिश्र संख्याऐं है, यदि $\text{Re}(z_1z_2)=0$ और $\text{Re}(z_1+z_2)=0$ है, तब

A. $\text{Im}(z_1)>0$ एवं $\text{Im}(z_2)>0$

B. $\text{Im}(z_1)<0$ एवं $\text{Im}(z_2)>0$

C. $\text{Im}(z_1)>0$ एवं $\text{Im}(z_2)<0$

D. $\text{Im}(z_1)<0$ एवं $\text{Im}(z_2)<0$

नीचे दिये गये विकल्पों से सही उत्तर चुनिए : **[JEE(Main)Jan.2023]**

(A) B तथा C (B) B तथा D (C) A तथा B (D) A तथा C

Ans. A

Sol. Let $z_1=x_1+iy_1$ and $z_2=x_2+iy_2$

$\because$ $\text{Re}(z_1z_2)=0$ and $\text{Re}(z_1+z_2)=0$,

$x_1x_2-y_1y_2=0$ ...(i)

$x_1 + x_2 = 0$ ...(ii)

$x_1^2 + y_1y_2 = 0$

$y_1y_2 = -x_1^2$

$\Rightarrow$ Im (z_1) and Im (z_2) are of opposite sign.

6. For all $z \in C$ on the curve $C_1 : |z| = 4$, let the locus of the point $z + \frac{1}{z}$ be the curve C_2. Then :

[JEE(Main)Jan.2023]

(A) the curves C_1 and C_2 intersect at 4 points

(B) the curve C_2 lies inside C_1

(C) the curve C_1 lies inside C_2

(D) the curves C_1 and C_2 intersect at 2 points

माना वक्र $C_1 : |z| = 4$ पर सभी $z \in C$ के लिए बिंदु $z + \frac{1}{z}$ का बिंदुपथ वक्र C_2 है, तो : **[JEE(Main)Jan.2023]**

(A) वक्र C_1 तथा C_2 चार बिंदुओं पर मिलते हैं

(B) वक्र C_2, वक्र C_1 के अन्दर है

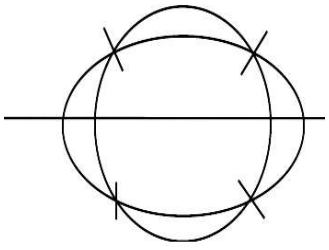
(C) वक्र C_1, वक्र C_2 के अन्दर है

(D) वक्र C_1 तथा C_2 दो बिंदुओं पर मिलते है

Ans. A

Sol. $|z| = 4$ is a circle $x^2 + y^2 = 16$ and $z + \frac{1}{z}$ is an ellipse $\frac{x^2}{\left(\frac{17}{4}\right)^2} + \frac{y^2}{\left(\frac{15}{4}\right)^2} = 1$

$\therefore$ the curves C_1 and C_2 intersect at 4 points



7. Let $z = 1 + i$ and $z_1 = \frac{1 + i\bar{z}}{\bar{z}(1 - z) + \frac{1}{z}}$. Then $\frac{12}{\pi}\arg(z_1)$ is equal to _____. **[JEE(Main)Jan.2023]**

माना $z = 1 + i$ तथा $z_1 = \frac{1 + i\bar{z}}{\bar{z}(1 - z) + \frac{1}{z}}$. है तो $\frac{12}{\pi}\arg(z_1)$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 9

Sol. $z = 1 + i$

$$z_1 = \frac{1 + i\bar{z}}{\bar{z}(1-z) + \frac{1}{z}}$$

$$z_1 = \frac{1 + i(1-i)}{(1-i)(1-1-i) + \frac{1}{1+I}}$$

$$= \frac{1 + i - i^2}{(1-i)(-i) + \frac{1-i}{2}}$$

$$= \frac{2+i}{\frac{-3i-1}{2}} = \frac{4+2i}{-3i-1}$$

$$= \frac{-(4+2i)(3i-1)}{(3i)^2 - (1)^2}$$

$$\text{Arg}(z_1) = \frac{3\pi}{4}$$

$$\therefore \frac{12}{\pi}\arg(z_1) = \frac{12}{\pi} \times \frac{3\pi}{4} = 9$$

8. The complex number $z = \dfrac{i-1}{\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}}$ is equal to : **[JEE(Main)Jan.2023]**

सम्मिश्र संख्या $z = \dfrac{i-1}{\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}}$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $\cos\frac{\pi}{12} - i\sin\frac{\pi}{12}$ (B) $\sqrt{2}i\left(\cos\frac{5\pi}{12} - i\sin\frac{5\pi}{12}\right)$

(C) $\sqrt{2}\left(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\right)$ (D) $\sqrt{2}\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)$

Ans. C

Sol. $$Z = \frac{i-1}{\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}} = \frac{\sqrt{2}\,e^{i3\pi/4}}{e^{i\pi/3}} = \sqrt{2}\,e^{i5\pi/12} = \sqrt{2}\left(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\right)$$

9. If the center and radius of the circle $\left|\frac{z-2}{z-3}\right|=2$ are respectively (α, β) and γ, then $3(\alpha+\beta+\gamma)$ is equal to :

[JEE(Main)Jan.2023]

यदि वृत्त $\left|\frac{z-2}{z-3}\right|=2$ के केन्द्र एवं त्रिज्या क्रमशः (α, β) एवं γ है, तब $3(\alpha+\beta+\gamma)$ का मान है :

[JEE(Main)Jan.2023]

(A) 9 (B) 11 (C) 10 (D) 12

Ans. D

Sol. $\left|\frac{z-2}{z-3}\right|=2$

$(z-2)(\overline{z}-2)=4(\overline{z}-3)(z-3)$

$\Rightarrow z\overline{z}-2z-2\overline{z}+4=4\left(z\overline{z}-3z-3\overline{z}+9\right)$

$3z\overline{z}-10z-10\overline{z}+32=0$

$z\overline{z}-\frac{10}{3}z-\frac{10}{3}\overline{z}+\frac{32}{3}=0$

$\alpha=-\frac{10}{3}$ $\quad$ $c=\frac{32}{3}$

centre $\left(\frac{10}{3},0\right)$ $\quad$ $\gamma=\sqrt{|a|^2-c}=\sqrt{\frac{100}{9}-\frac{32}{3}}=\frac{2}{3}$

$\alpha=\frac{10}{3}, \beta=0, \gamma=\frac{2}{3}$

$\therefore \alpha+\beta+\gamma=\frac{10}{3}+\frac{2}{3}=\frac{12}{3}$ $\quad$ $\therefore 3(\alpha+\beta+\psi)=12$

10. Let a, b be two real numbers such that $ab<0$. If the complex number $\frac{1+ai}{b+i}$ is of unit modulus and $a+ib$ lies on the circle $|z-1|=|2z|$, then a possible value of $\frac{1+[a]}{4b}$, where $[t]$ is greatest integer :

[JEE(Main)Jan.2023]

माना a, b दो वास्तविक संख्याऐं है, जिन के लिए $ab<0$ है। यदि सम्मिश्र संख्या $\frac{1+ai}{b+i}$ इकाई मापक की है तथा $a+ib$ वृत्त, $|z-1|=|2z|$ पर है, तो $\frac{1+[a]}{4b}$ जहाँ $[t]$ महत्तम पूर्णांक फलन है, का एक संभव मान है : **[JEE(Main)Jan.2023]**

(A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) 1 (D) –1

Ans. Drop

Sol. $ab < 0, \left|\frac{1+ai}{b+i}\right| = 1$

$|1 + ai| = |b + i|$

$a^2 + 1 = b^2 + 1 \Rightarrow a = \pm b \Rightarrow b = -a$ as $ab < 0$

(a, b) lies on $|z - 1| = |2z|$

$|a + ib - 1| = 2|a + ib|$

$(a - 1)^2 + b^2 = 4(a^2 + b^2)$

$(a - 1)^2 + a^2 = 4(2a^2)$

$1 - 2a = 6a^2 \Rightarrow 6a^2 + 2a - 1 = 0$

$$a = \frac{-2 \pm \sqrt{28}}{12} = \frac{-1 \pm \sqrt{7}}{6}$$

$$a = \frac{\sqrt{7} - 1}{6} \ \& \ b = \frac{1 - \sqrt{7}}{6}$$

$[a] = 0$

$$\therefore \frac{1 + [a]}{4b} = \frac{6}{4\left(1 - \sqrt{7}\right)} = -\left(\frac{1 + \sqrt{7}}{4}\right)$$

or $[a] = 0$

Similarly it is not matching with $a = \frac{-1 - \sqrt{7}}{6}$

No answer is matching.

11. Let $\alpha = 8 - 14i$, $A = \left\{z \in C : \frac{\alpha z - \bar{\alpha}\bar{z}}{z^2 - (\bar{z})^2 - 112i} = 1\right\}$ and $B = \{z \in C : |z + 3i| = 4\}$. Then $\sum_{z \in A \cap B} (\text{Re}\, z - \text{Im}\, z)$ is equal to _____. **[JEE(Main)Jan.2023]**

माना $\alpha = 8 - 14i$, $A = \left\{z \in C : \frac{\alpha z - \bar{\alpha}\bar{z}}{z^2 - (\bar{z})^2 - 112i} = 1\right\}$ तथा $B = \{z \in C : |z + 3i| = 4\}$ है। तो $\sum_{z \in A \cap B} (\text{Re}\, z - \text{Im}\, z)$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 14

Sol. $\alpha = 8 - 14i$

$z = x + iy$

$az = (8x + 14y) + i(-14x + 8y)$

$z + \bar{z} = 2x$ $z - \bar{z} = 2iy$

$$\text{Set A}: \frac{2i(-14x + 8y)}{i(4xy - 112)} = 1$$

$(x - 4)(y + 7) = 0$

$x = 4$ or $y = -7$

Set B : $x^2 + (y + 3)^2 = 16$

when $x = 4$ $y = -3$

when $y = -7$ $\quad x = 0$

$\therefore A \cap B = \{4 - 3i, 0 - 7i\}$

So, $\sum_{z \in A \cap B} (\text{Re}\, z - \text{Im}\, z) = 4 - (-3) + (0 - (-7)) = 14$

12. Let $a \neq b$ be two non-zero real numbers. Then the number of elements in the set $X = \left\{z \in \mathbb{C} : \text{Re}\left(az^2 + bz\right) = a \text{ and } \text{Re}\left(bz^2 + az\right) = b\right\}$ is equal to : **[JEE(Main)April,2023]**

माना $a \neq b$ दो शून्येतर वास्तविक संख्याएँ है। तो समुच्चय $X = \left\{z \in \mathbb{C} : \text{Re}\left(az^2 + bz\right) = a \text{ तथा } \text{Re}\left(bz^2 + az\right) = b\right\}$ में

v o; okLrfod hl la[;k gS **[JEE(Main)April,2023]**

(A) 1 (B) 3 (C) 2 (D) 0

Ans. D

Sol. $\text{Re}\left(az^2 + bz\right) = a$

$az^2 + bz + a\bar{z}^2 + b\bar{z} = 2a$

$a\left(z^2 + \bar{z}^2\right) + b(z + \bar{z}) = 2a$ .....(1)

$\text{Re}\left(bz^2 + az\right) = b$

$bz^2 + az + b\bar{z}^2 + a\bar{z} = 2b$

$b\left(z^2 + \bar{z}^2\right) + a(z + \bar{z}) = 2b$ .....(2)

$(1) \times b - (2) \times (a)$

$\Rightarrow \left(b^2 - a^2\right)(z + \bar{z}) = 0$

$\Rightarrow \quad (z + \bar{z}) = 0 \quad \left(a^2 \neq b^2\right)$

$(1) \times a - (2) \times (b)$

$\Rightarrow \quad \left(a^2 - b^2\right)(z + \bar{z}) = 2\left(a^2 - b^2\right) \quad \left(a^2 \neq b^2\right)$

$z^2 + \bar{z}^2 = 2$

$\Rightarrow (z + \bar{z})^2 - 2z\bar{z} = 2$

$z\bar{z} = -1$

$\Rightarrow 1 + 1^2 = -1$

$\Rightarrow$ No solution

But when $a = -b$,

$\text{Re}(az - az) = a$

$\Rightarrow \quad \text{Re}\left(a\left(x^2 - y^2 + i2xy\right) - a(x + iy)\right) = a$

$\Rightarrow a\left(x^2 - y^2\right) - ax = a$

$\Rightarrow x^2 - y^2 - x = 1$

$\Rightarrow x^2 - x - 1 = y^2$

For any real values of y there two values of x, hence infinite complex numbers are possible.

13. If for $z = \alpha + i\beta, |z+2| = z + 4(1+i)$, then $\alpha + \beta$ and $\alpha\beta$ are the roots of the equation :

[JEE(Main)April,2023]

यदि $z = \alpha + i\beta$ के लिए $|z+2| = z + 4(1+i)$, तो $\alpha + \beta$ तथा $\alpha\beta$ किस समीकरण के मूल है ?

[JEE(Main)April,2023]

(A) $x^2 + 7x + 12 = 0$ (B) $x^2 + 2x - 3 = 0$ (C) $x^2 + x - 12 = 0$ (D) $x^2 + 3x - 4 = 0$

Ans. A

Sol. $z = \alpha + i\beta$

$|z+2| = z + 4(1+i)$

$|(\alpha+2) + i\beta| = \alpha + i\beta + 4 + 4i$

$\sqrt{(\alpha+2)^2 + \beta^2} = (\alpha+4) + i(\beta+4)$

compare real and imaginary part from both sides

$\beta + 4 = 0 \Rightarrow \beta = -4$

and $\sqrt{(\alpha+2)^2 + 16} = \alpha + 4$

$\Rightarrow \alpha^2 + 4 + 4\alpha + 16 = \alpha^2 + 16 + 8\alpha$

$\Rightarrow \alpha = 1, \beta = -4$

$\alpha + \beta = -3, \alpha\beta = -4$

Hence equation is $x^2 - (-3-4)x + (-3)(-4) = 0$

$x^2 + 7x + 12 = 0$

14. Let $A = \left\{\theta \in (0, 2\pi) : \frac{1 + 2i\sin\theta}{1 - i\sin\theta} \text{ is purely imaginary}\right\}$. Then the sum of the elements in A is :

[JEE(Main)April,2023]

माना $A = \left\{\theta \in (0, 2\pi) : \frac{1 + 2i\sin\theta}{1 - i\sin\theta} \text{ मात्र काल्पनिक है}\right\}$। तो A में अवयवों का योग है : **[JEE(Main)April,2023]**

(A) π (B) 4π (C) 2π (D) 3π

Ans. B

Sol. $\frac{1 + 2i\sin\theta}{1 - i\sin\theta}$ is purely imaginary

So, $\frac{1 - 2\sin^2\theta}{1 + \sin^2\theta} = 0$

$\Rightarrow \sin^2\theta = \frac{1}{2}$

$\therefore \quad \theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \text{ for } \theta \in (0, 2\pi)$

$\therefore \quad$ Sum of all values $= \frac{16\pi}{4} = 4\pi$

15. Let the complex number $z = x + iy$ be such that $\frac{2z-3i}{2z+i}$ is purely imaginary. If $x + y^2 = 0$, then $y^4 + y^2 - y$ is equal to : **[JEE(Main)April,2023]**

माना सम्मिश्र संख्या $z = x + iy$ के लिए $\frac{2z-3i}{2z+i}$ मात्र काल्पनिक है। यदि $x + y^2 = 0$ है, तो $y^4 + y^2 - y$ बराबर है : **[JEE(Main)April,2023]**

(A) $\frac{4}{3}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{3}{2}$

Ans. C

Sol. $z = x + iy$

$$\frac{(2z-3i)}{2z+i} = \text{purely imaginary}$$

$$\text{means } \text{Re}\left(\frac{2z-3i}{2z+i}\right) = 0$$

$$\Rightarrow \frac{(2z-3i)}{(2z+i)} = \frac{2(x+iy)-3i}{2(x+iy)+i}$$

$$= \frac{2x+2yi-3i}{2x+i2y+i}$$

$$= \frac{2x+i(2y-3)}{2x+i(2y+1)} \times \frac{2x-i(2y+1)}{2x-i(2y+1)}$$

$$\text{Re}\left[\frac{2z-3i}{2z+i}\right] = \frac{4x^2+(2y-3)(2y+1)}{4x^2+(2y+1)^2} = 0$$

$$\Rightarrow 4x^2+(2y-3)(2y+1) = 0$$

$$\Rightarrow 4x^2+\left[4y^2+2y-6y-3\right] = 0$$

$$\because x + y^2 = 0 \Rightarrow x = -y^2$$

$$\Rightarrow 4\left(-y^2\right)^2 + 4y^2 - 4y - 3 = 0$$

$$\Rightarrow 4y^4 + 4y^2 - 4y = 3$$

$$\Rightarrow y^4 + y^2 - y = \frac{3}{4}$$

Therefore, correct answer is option (3).

16. Let $S=\left\{z=x+iy:\frac{2z-3i}{4z+2i} \text{ is a real number}\right\}$. Then which of the following is NOT correct :

[JEE(Main)April,2023]

माना $S=\left\{z=x+iy:\frac{2z-3i}{4z+2i} \text{ एक वास्तविक संख्या}\right\}$। तो निम्न में कौन सा सही नहीं है ? **[JEE(Main)April,2023]**

(A) $y+x^2+y^2 \neq -\frac{1}{4}$ (B) $x=0$

(C) $(x,y)=\left(0,-\frac{1}{2}\right)$ (D) $y \in \left(-\infty,-\frac{1}{2}\right)\cup\left(-\frac{1}{2},\infty\right)$

Ans. C

Sol. $\because$ $z=x+iy$ and $\frac{2z-3i}{4z+2i}$ is real number

$\therefore$ $\frac{2x+(2y-3)i}{4x+(4y+2)i}$ is real number.

$\therefore$ $\frac{(2x+(2y-3)i)(4x-(4y+2)i)}{(4x)^2+(4y+2)^2}$ is real number

$\therefore$ $-2x(4y+2)+(2y-3)4x=0$

$\therefore$ $x=0$

Here $x=0$ and $y \in \mathbb{R}$ but $y \neq -\frac{1}{2}$ is not acceptable

in this case

Denominator will be zero.

17. Let w_1 be the point obtained by the rotation of $z_1=5+4i$ about the origin through a right angle in the anticlockwise direction, and w_2 be the point obtained by the rotation of $z_2=3+5i$ about the origin through a right angle in the clockwise direction. Then the principal argument of w_1-w_2 is equal to : **[JEE(Main)April,2023]**

माना $z_1=5+4i$ को मूल बिंदु के सापेक्ष घड़ी की विपरीत दिशा में एक समकोण तक घुमाने पर बिंदु w_1 प्राप्त होता है तथा $z_2=3+5i$ को मूलबिंदु के सापेक्ष घड़ी की दिशा में एक समकोण तक घुमाने पर बिंदु w_2 प्राप्त होता है। तो w_1-w_2 का मुख्य आयाम बराबर है :

[JEE(Main)April,2023]

(A) $-\pi+\tan^{-1}\frac{8}{9}$ (B) $\pi-\tan^{-1}\frac{33}{5}$ (C) $-\pi+\tan^{-1}\frac{33}{5}$ (D) $\pi-\tan^{-1}\frac{8}{9}$

Ans. D

Sol. $W_1=z_i=(5+4i)i=-4+5i\ldots$ (i)

$W_1=z_2(-i)=(3+5i)(-i)=5-3i\ldots(2)$

$W_1-W_2=-9+8i$

Principal argument $= \pi - \tan^{-1}\left(\frac{8}{9}\right)$

18. For $a \in \mathbb{C}$, let $A = \{z \in \mathbb{C} : Re(a + \overline{z}) > Im(\overline{a} + z)\}$ and $B = \{z \in \mathbb{C} : Re(a + \overline{z}) < Im(\overline{a} + z)\}$. Then among the two statements :

(S1) : If Re (a), Im (a) > 0, then the set A contains all the real numbers

(S2) : If Re (a), Im (a) < 0, then the set B contains all the real numbers **[JEE(Main)April,2023]**

(A) only (S1) is true (B) both are false (C) only (S2) is true (D) both are true

$a \in \mathbb{C}$ के लिए, माना $A = \{z \in \mathbb{C} : Re(a + \overline{z}) > Im(\overline{a} + z)\}$ तथा $B = \{z \in \mathbb{C} : Re(a + \overline{z}) < Im(\overline{a} + z)\}$ हैं। तो दो कथनों:

(S1) : यदि Re (a), Im (a) > 0 है, तो सभी वास्तविक संख्याएँ A में हैं

(S2) : यदि Re (a), Im (a) < 0 है, तो सभी वास्तविक संख्याएँ B में हैं **[JEE(Main)April,2023]**

(A) केवल (S1) सही है (B) दोनो गलत है (C) केवल (S2) सही है (D) दोनो सही है

Ans. B

Sol. Let $a = x_1 + iy_1, z = x + iy$

Now $Re(a + \overline{z}) > Im(\overline{a} + z)$

$\therefore x_1 + x > -y_1 + y$

$x_1 = 2, y_1 = 10, x = -12, y = 0$

Given inequality is not valid for these values.

S1 is false.

Now $Re(a + \overline{z}) < Im(\overline{a} + z)$

$x_1 + x < -y_1 + y$

$x_1 = -2, y_1 = -10, x = 12, y = 0$

Given inequality is not valid for these values.

S2 is false.

19. Let C be the circle in the complex plane with centre $z_0 = \frac{1}{2}(1 + 3i)$ and radius r = 1. Let $z_1 = 1 + i$ and the complex number z_2 be outside the circle C such that $|z_1 - z_0||z_2 - z_0| = 1$. If z_0, z_1 and z_2 are collinear, then the smaller value of $|z_2|^2$ is equal to : **[JEE(Main)April,2023]**

माना सम्मिश्र तल में वृत C का केन्द्र $z_0 = \frac{1}{2}(1 + 3i)$ तथा त्रिज्या r = 1 हैं। माना $z_1 = 1 + i$ है तथा वृत्त के बाहर सम्मिश्र संख्या z_2 इस प्रकार है कि $|z_1 - z_0||z_2 - z_0| = 1$ है। यदि z_0, z_1 तथा z_2 संरेख हैं, तो $|z_2|^2$ का छोटा मान बराबर है :

[JEE(Main)April,2023]

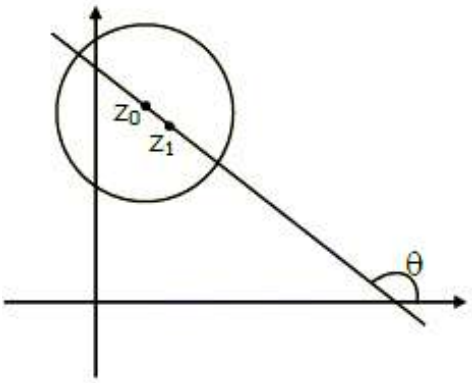
(A) $\frac{7}{2}$ (B) $\frac{13}{2}$ (C) $\frac{3}{2}$ (D) $\frac{5}{2}$

Ans. D

Sol. $|z_1 - z_0| = \left|\frac{1-i}{2}\right| = \frac{1}{\sqrt{2}}$

$\Rightarrow |z_2 - z_\circ| = \sqrt{2};$ centre $\left(\frac{1}{2}, \frac{3}{2}\right)$

$z_\circ\left(\frac{1}{2}, \frac{3}{2}\right)$ and $z_1(1,1)$



$\tan\theta = -1 \Rightarrow \theta = 135^\circ$

$z_2\left(\frac{1}{2} + \sqrt{2}\cos 135^\circ, \frac{3}{2} + \sqrt{2}\sin 135^\circ\right)$

or

$\left(\frac{1}{2} - \sqrt{2}\cos 135^\circ, \frac{3}{2} - \sqrt{2}\sin 135^\circ\right)$

$\Rightarrow z_2\left(-\frac{1}{2}, \frac{5}{2}\right)$ or $z_2\left(\frac{3}{2}, \frac{1}{2}\right)$

$\Rightarrow |z_2|^2 = \frac{26}{4}, \frac{5}{2}$

$\Rightarrow |z_2|_{\min^2} = \frac{5}{2}$

20. Let α, β be the roots of the equation $x^2 - \sqrt{2}x + 2 = 0$. Then $\alpha^{14} + \beta^{14}$ is equal to :

[JEE(Main)April,2023]

माना समीकरण $x^2 - \sqrt{2}x + 2 = 0$ के मूल α, β हैं। तो $\alpha^{14} + \beta^{14}$ बराबर है : **[JEE(Main)April,2023]**

(A) -64 (B) -128 (C) $-64\sqrt{2}$ (D) $-128\sqrt{2}$

Ans. B

Sol. $x^2 - \sqrt{2}x + 2 = 0$

$x = \frac{\sqrt{2} \pm \sqrt{2-8}}{2} = \frac{\sqrt{2} \pm \sqrt{6}i}{2}$

$\alpha = \frac{\sqrt{2} + \sqrt{6}i}{2} = \sqrt{2}e^{\frac{i\pi}{3}}$ & $\beta = \sqrt{2}e^{\frac{-i\pi}{3}}$

$\alpha^{14} = 2^7 e^{\frac{i14\pi}{3}} = 128\left[e^{\frac{i2\pi}{3}}\right]$

$\beta^{14} = 128\left[e^{\frac{-i2\pi}{3}}\right]$

$$\alpha^{14}+\beta^{14}=128(2)\cos\left(\frac{2\pi}{3}\right)=-128$$

21. Let $S=\left\{z\in\mathbb{C}:\overline{z}=i\left(z^2+\text{Re}(\overline{z})\right)\right\}$. The $\sum_{z\in S}|z|^2$ is equal to : **[JEE(Main)April,2023]**

माना $S=\left\{z\in\mathbb{C}:\overline{z}=i\left(z^2+\text{Re}(\overline{z})\right)\right\}$ है। तो $\sum_{z\in S}|z|^2$ बराबर है : **[JEE(Main)April,2023]**

(A) $\frac{5}{2}$ (B) 4 (C) 3 (D) $\frac{7}{2}$

Ans. B

Sol. Let $z = x + iy$

$\overline{z}=i\left(z^2+\text{Re}\left(\overline{z}\right)\right)$

$\Rightarrow x - iy = i(x^2 - y^2 + 2i\,xy + x)$

$x - iy = i(x^2 - y^2 + x) - 2xy$

$x = -2xy \Rightarrow x(2y + 1) = 0$

$\Rightarrow x = 0, y = \frac{-1}{2}$ ....(i)

$-y = x^2 - y^2 + x$ ....(ii)

Case (1) $x = 0$

(ii) $\Rightarrow -y = -y^2 \Rightarrow y^2 - y = 0 \Rightarrow y = 0, 1$

$z = 0, i$

Case (2) $y = \frac{-1}{2}$

(ii) $\Rightarrow \frac{1}{2} = x^2 - \frac{1}{4} + x \Rightarrow x^2 + x - \frac{3}{4} = 0$

$4x + 4x - 3 = 0 \Rightarrow (2x - 1)(2x + 3) = 0$

$x = \frac{1}{2}, \frac{-3}{2}$

$z = \frac{1}{2} - \frac{1}{2}i, \frac{-3}{2} - \frac{1}{2}i$

$\sum|z|^2 = 0 + 1 + \frac{1}{2} + \frac{5}{2} = 4$

22. If the set $\left\{\text{Re}\left(\frac{z-\overline{z}+z\overline{z}}{2-3z+5\overline{z}}\right):z\in\mathbb{C},\text{Re}(z)=3\right\}$ is equal to the interval $(\alpha, \beta]$, then $24(\beta - \alpha)$ is equal to :

[JEE(Main)April,2023]

यदि समुच्चय $\left\{\text{Re}\left(\frac{z-\overline{z}+z\overline{z}}{2-3z+5\overline{z}}\right):z\in\mathbb{C},\text{Re}(z)=3\right\}$ अन्तराल $(\alpha, \beta]$ के बराबर है, तो $24(\beta - \alpha)$ का मान होगा :

(A) 27 (B) 42 (C) 36 (D) 30

Ans. D

Sol. Let $z_1 = \left(\frac{z-\overline{z}+z\overline{z}}{2-3z+5\overline{z}}\right)$

Let $z = 3 + iy$

$\overline{z} = 3 - iy$

$$z_1 = \frac{2iy + \left(9+y^2\right)}{2-3\left(3+iy\right)+5\left(3-iy\right)}$$

$$= \frac{9+y^2+i(2y)}{8-8iy}$$

$$= \frac{\left(9+y^2\right)+i(2y)}{8(1-iy)}$$

$$\mathrm{Re}(z_1) = \frac{\left(9+y^2\right)-2y^2}{8\left(1+y^2\right)}$$

$$= \frac{9-y^2}{8\left(1+y^2\right)}$$

$$= \frac{1}{8}\left[\frac{10-\left(1+y^2\right)}{\left(1+y^2\right)}\right]$$

$$= \frac{1}{8}\left[\frac{10}{1+y^2}-1\right]$$

$1 + y^2 \in [1, \infty]$

$$\frac{1}{1+y^2} \in (0,1]$$

$$\frac{10}{1+y^2} \in (0,10]$$

$$\frac{10}{1+y^2}-1 \in (-1,9]$$

$$\mathrm{Re}(z_1) \in \left(\frac{-1}{8}, \frac{9}{8}\right]$$

$\alpha = \frac{-1}{8}, \beta = \frac{9}{8}$

$24(\beta - \alpha) = 24\left(\frac{9}{8} + \frac{1}{8}\right) = 30$

23. For $\alpha, \beta, z \in \mathbb{C}$ and $\lambda > 1$, If $\sqrt{\lambda - 1}$ is the radius of the circle $|z - \alpha|^2 + |z - \beta|^2 = 2\lambda$, then $|\alpha - \beta|$ is equal to _____. **[JEE(Main)April,2023]**

$\alpha, \beta, z \in \mathbb{C}$ तथा $\lambda > 1$ के लिए, यदि वृत्त $|z - \alpha|^2 + |z - \beta|^2 = 2\lambda$ की त्रिज्या $\sqrt{\lambda - 1}$ है, तो $|\alpha - \beta|$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 2

Sol. $|z - z_1|^2 + |z - z_2|^2 = |z_1 - z_2|^2$

$z_1 = \alpha, z_2 = \beta$

$|\alpha - \beta|^2 = 2\lambda$

$|\alpha - \beta| = \sqrt{2\lambda}$

$2r = \sqrt{2\lambda}$

$2\sqrt{\lambda - 1} = \sqrt{2\lambda}$

$\Rightarrow 4(\lambda - 1) = 2\lambda$

$\lambda = 2$

$|\alpha - \beta = 2$

24. Let $S = \left\{ z \in \mathbb{C} - \{i, 2i\} : \frac{z^2 + 8iz - 15}{z^2 - 3iz - 2} \in \mathbb{R} \right\}$. If $\alpha - \frac{13}{11}i \in S, \alpha \in \mathbb{R} - \{0\}$, then $242\alpha^2$ is equal to _____. **[JEE(Main)April,2023]**

माना $S = \left\{ z \in \mathbb{C} - \{i, 2i\} : \frac{z^2 + 8iz - 15}{z^2 - 3iz - 2} \in \mathbb{R} \right\}$ है। यदि $\alpha - \frac{13}{11}i \in S, \alpha \in \mathbb{R} - \{0\}$ है, तो $242\alpha^2$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 1680

Sol. $\left(\frac{z^2 + 8iz - 15}{z^2 - 3iz - 2}\right) \in R$

$\Rightarrow 1 + \frac{(1 \text{ liz } -13)}{\left(z^2 - 3iz - 2\right)} \in R$

Put $z = \alpha - \frac{13}{11}i$

$\Rightarrow \left(z^2 - 3iz - 2\right)$ is imaginary

Put $z = x + iy$

$\Rightarrow \left(x^2 - y^2 + 2xyi - 3ix + 3y - 2\right) \in$ Imaginary

$\Rightarrow \text{Re}\left(x^2 - y^2 + 3y - 2 + (2xy - 3x)i\right) = 0$

$\Rightarrow x^2 - y^2 + 3y - 2 = 0$

$x^2 = y^2 - 3y + 2$

$y^2 = (y - 1)(y - 2)$

$\therefore z = \alpha - \frac{13}{11}i$

Put $x = \alpha,\ y = \frac{-13}{11}$

$\alpha^2 = \left(\frac{-13}{11} - 1\right)\left(\frac{-13}{11} - 2\right)$

$\alpha^2 = \frac{(24 \times 35)}{121}$

$242\alpha^2 = 48 \times 35 = 1680$

25. Let $w = z\bar{z} + k_1 z + k_2 iz + \lambda(1+i), k_1, k_2 \in \mathbb{R}$. Let Re(w) = 0 be the circle C of radius 1 in the first quadrant touching the line y = 1 and the y-axis. If the curve Im(w) = 0 intersects C at A and B, then $30(AB)^2$ is equal to _____. **[JEE(Main)April,2023]**

माना $w = z\bar{z} + k_1 z + k_2 iz + \lambda(1+i), k_1, k_2 \in \mathbb{R}$ है। माना Re(w) = 0 प्रथम चतुर्थांश में इकाई त्रिज्या का एक वृत्त C है, जो रेखा y = 1 तथा y-अक्ष को स्पर्श करता है। यदि वक्र Im(w) = 0, वृत्त C को A तथा B पर काटता है, तो $30(AB)^2$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 24

Sol. $\omega = z\bar{z} + k_1 z + k_2 iz + \lambda(1+i)$

$\text{Re}(\omega) = x^2 + y^2 + k_1 x - x_2 y + \lambda = 0$

$\text{Centre} \equiv \left(\frac{-k_1}{2}, \frac{k_2}{2}\right) \equiv (1, 2)$

$\Rightarrow k_1 = -2,\ k_2 = 4$

$\rho\omega\delta\omega\sigma = 1 \Rightarrow \lambda - 4 = 0$

$\text{Im} = k_1 y + k_2 x + \lambda = 0$

$\therefore\ 2x - y + 2 = 0$

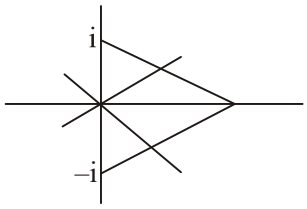
26. If $S = \{z \in C : |z - i| = |z + i| = |z - 1|\}$, then n(S) is : **[JEE(Main)Jan.2024]**

यदि If $S = \{z \in C : |z - i| = |z + i| = |z - 1|\}$ है, तो n(S) बराबर है : **[JEE(Main)Jan.2024]**

(A) 2 (B) 0 (C) 3 (D) 1

Ans. D

Sol.



27. If $z=\frac{1}{2}-2i$ is such that $|z+1|=\alpha z+\beta(1+i), i=\sqrt{-1}$ and $\alpha,\beta\in R$, then $\alpha+\beta$ is :

[JEE(Main)Jan.2024]

यदि $z=\frac{1}{2}-2i$ के लिए $|z+1|=\alpha z+\beta(1+i)$ हैं, जहाँ $i=\sqrt{-1}$ तथा $\alpha,\beta\in R$ है, तो $\alpha+\beta$ बराबर है :

[JEE(Main)Jan.2024]

(A) 2 (B) – 4 (C) 3 (D) –1

Ans. C

Sol. $\left|\frac{3}{2}-2i\right|=\left(\frac{\alpha}{2}-2\alpha i\right)+\beta i+\beta$

$2\alpha=\beta$ and $\frac{\alpha}{2}+\beta=\sqrt{\frac{9}{4}+4}$

$\frac{\alpha}{2}+2\alpha=\frac{5}{2}$

$\alpha = 1$

$\beta = 2$

$|\alpha+\beta|=3$

28. Let r and θ respectively be the modulus and amplitude of the complex number $z=2-i\left(2\tan\frac{5\pi}{8}\right)$, then (r, θ) is equal to :

[JEE(Main)Jan.2024]

यदि सम्मिश्र संख्या $z=2-i\left(2\tan\frac{5\pi}{8}\right)$ के मापांक तथा कोणांक क्रमशः r तथा θ हैं, तो (r, θ) बराबर है :

[JEE(Main)Jan.2024]

(A) $\left(2\sec\frac{11\pi}{8},\frac{11\pi}{8}\right)$ (B) $\left(2\sec\frac{3\pi}{8},\frac{5\pi}{8}\right)$ (C) $\left(2\sec\frac{3\pi}{8},\frac{3\pi}{8}\right)$ (D) $\left(2\sec\frac{5\pi}{8},\frac{3\pi}{8}\right)$

Ans. C

Sol. $z = 2 - i\left(2\tan\frac{5\pi}{8}\right) = x + iy$ (let)

$r = \sqrt{x^2 + y^2}$

$\theta = \tan^{-1}\frac{y}{x}$

$r = \sqrt{(2)^2 + \left(2\tan\frac{5\pi}{8}\right)^2}$

$= \left|2\sec\frac{5\pi}{8}\right| = \left|2\sec\left(\pi - \frac{3\pi}{8}\right)\right|$

$= 2\sec\frac{3\pi}{8}$

$\theta = \tan^{-1}$

$\frac{-2\tan\frac{5\pi}{8}}{2}$

$= \tan^{-1}\left(\tan\left(\pi - \frac{5\pi}{8}\right)\right) = \frac{3\pi}{8}$

29. If $z = x + iy$, $xy \neq 0$, satisfies the equation $z^2 + i\bar{z} = 0$, then $|z^2|$ is equal to : **[JEE(Main)Jan.2024]**

यदि $z = x + iy$, $xy \neq 0$ समीकरण $z^2 + i\bar{z} = 0$ को संतुष्ट करता है, तो $|z^2|$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 1 (B) 1/4 (C) 9 (D) 4

Ans. A

Sol. $z = x + iy$, $x \neq 0$, $y \neq 0$ $\Rightarrow |z| \neq 0$

$z^2 + i\bar{z} = 0$

$\Rightarrow z^2 = -i\bar{z}$

$\Rightarrow \left|z^2\right| = \left|-i\bar{z}\right| = \left|-i\right|\left|\bar{z}\right|$

$\Rightarrow |z|^2 = |z| \Rightarrow |z|(|z| - 1) = 0$

$\Rightarrow |z| = 1$

30. If z is a complex number, then the number of common roots of the equations $z^{1985} + z^{100} + 1 = 0$ and $z^3 + 2z^2 + 2z + 1 = 0$, is equal to : **[JEE(Main)Jan.2024]**

यदि z एक सम्मिश्र संख्या हैं, तो समीकरणों $z^{1985} + z^{100} + 1 = 0$ तथा $z^3 + 2z^2 + 2z + 1 = 0$ के उभयनिष्ठों मूलों की संख्या है : **[JEE(Main)Jan.2024]**

(A) 0 (B) 3 (C) 1 (D) 2

Ans. D

Sol. Given $z^{1985} + z^{100} + 1 = 0$ and $z^3 + 2z^2 + 2z + 1 = 0$

The roots of $z^{1985} + z^{100} + 1 = 0$ be ω and ω^2 and also satisfies $z^3 + 2z^2 + 2z + 1 = 0$

$\therefore$ ω and ω^2 are common solutions

(where ω is cube root of unity)

$\therefore$ 2 solutions

31. Let z_1 and z_2 be two complex numbers such that $z_1 + z_2 = 5$ and $z_1^3 + z_2^3 = 20 + 15i$. Then $|z_1^4 + z_2^4|$ equals :

[JEE(Main)Jan.2024]

माना दो साम्मिश्र संख्याओं z_1 तथा z_2 के लिए $z_1 + z_2 = 5$ तथा $z_1^3 + z_2^3 = 20 + 15i$ हैं। तो $|z_1^4 + z_2^4|$ बराबर है :

[JEE(Main)Jan.2024]

(A) $30\sqrt{3}$ (B) $25\sqrt{3}$ (C) $15\sqrt{15}$ (D) 75

Ans. D

Sol. $z_1 + z_2 = 5$

$z_1^3 + z_2^3 = 20 + 15i$

$z_1^3 + z_2^3 = (z_1 + z_2)^3 - 3z_1z_2(z_1 + z_2)$

$z_1^3 + z_2^3 = 125 - 3z_1.z_2(5)$

$\Rightarrow 20 + 15i = 125 - 15z_1z_2$

$\Rightarrow 3z_1z_2 = 25 - 4 - 3i$

$\Rightarrow 3z_1z_2 = 21 - 3i$

$\Rightarrow z_1 \cdot z_2 = 7 - i$

$\Rightarrow (z_1 + z_2)^2 = 25$

$\Rightarrow z_1^2 + z_2^2 = 25 - 2(7 - i)$

$\Rightarrow 11 + 2i$

$(z_1^2 + z_2^2)^2 = 121 - 4 + 44i$

$\Rightarrow z_1^4 + z_2^4 + 2(7 - i)^2 = 117 + 44i$

$\Rightarrow z_1^4 + z_2^4 = 117 + 44i - 2(49 - 1 - 14i)$

$\Rightarrow |z_1^4 + z_2^4| = 75$

32. Let $S = \{z \in C : |z - 1| = 1$ and $(\sqrt{2} - 1)(z + \bar{z}) - i(z - \bar{z}) = 2\sqrt{2}\}$. Let $z_1, z_2 \in S$ be such that $|z_1| = \max_{Z \in S} |z|$ and $|z_2| = \min_{z \in S} |z|$. Then $|\sqrt{2}z_1 - z_2|^2$ equals : **[JEE(Main)Jan.2024]**

माना $S = \{z \in C : |z-1| = 1$ तथा $\left(\sqrt{2}-1\right)\left(z+\bar{z}\right)-i\left(z-\bar{z}\right)=2\sqrt{2}\}$ है। माना $z_1, z_2 \in S$ के लिए $|z_1| = \max_{Z\in S}|z|$ तथा $|z_2| = \min_{Z\in S}|z|$ हैं। तो $\left|\sqrt{2}z_1 - z_2\right|^2$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. $\left(\sqrt{2}-1\right)\left(z+\bar{z}\right)-i\left(z-\bar{z}\right)=2\sqrt{2}$

$\left(\sqrt{2}-1\right)\left(2x\right)-i\left(2iy\right)=2\sqrt{2}$

$\left(\sqrt{2}-1\right)x+y=\sqrt{2}$

and circle $|z-1|=1$

$(x-1)^2+y^2=1$

$\left(x-1\right)^2+\left(\sqrt{2}-\left(\sqrt{2}-1\right)x\right)^2=1$

$\left(x-1\right)^2+\left(\sqrt{2}\left(1-x\right)+x\right)^2=1$

$\left(x-1\right)^2+2\left(1-x\right)^2+x^2+2x\sqrt{2}\left(1-x\right)=1$

$3\left(x-1\right)^2+x^2+2\sqrt{2}x\left(1-x\right)=1$

$\left(-2\sqrt{2}+4\right)x^2+x\left(-6+2\sqrt{2}\right)+2=0$

$x = 1$ or $x=\dfrac{1}{2-\sqrt{2}}$

for $x=1 \Rightarrow y=1$

and $x=\dfrac{1}{2-\sqrt{2}} \Rightarrow y=\sqrt{2}-\dfrac{1}{\sqrt{2}}$

So, $\left|\sqrt{2}z_1-z_2\right|^2=\left|\dfrac{1}{\sqrt{2}-1}+i-1-i\right|^2=2$

33. If z is a complex number such that $|z| \geq 1$, then the minimum value of $\left|z+\dfrac{1}{2}\left(3+4i\right)\right|$ is :

[JEE(Main)Jan.2024]

यदि एक सम्मिश्र संख्या z के लिए $|z| \geq 1$ है, तो $\left|z+\dfrac{1}{2}\left(3+4i\right)\right|$ का न्यूनतम मान है : **[JEE(Main)Jan.2024]**

(A) 3 (B) $\dfrac{5}{2}$ (C) 2 (D) $\dfrac{3}{2}$

Ans. Bonus

Sol. p(z) is a point lying outside circle $x^2+y^2=1$

$Q\left(\frac{-3}{2},-2\right)$

$\left|z+\frac{1}{2}(3+4i)\right| = PQ$

$(PQ)_{min} = 0$ when P coincides with Q.

34. If α satisfies the equation $x^2 + x + 1 = 0$ and $(1+\alpha)^7 = A + B\alpha + C\alpha^2$, A, B, C ≥ 0, then $5(3A - 2B - C)$ is equal to _____. **[JEE(Main)Jan.2024]**

यदि α समीकरण $x^2 + x + 1 = 0$ को संतुष्ट करता है तथा $(1+\alpha)^7 = A + B\alpha + C\alpha^2$, A, B, C ≥ 0 हैं, तो $5(3A - 2B - C)$ बराबर है _____। **[JEE(Main)Jan.2024]**

Ans. 5

Sol. $\alpha^2 + \alpha + 1 = 0 \left\langle \begin{matrix} \omega \\ \omega^2 \end{matrix} \right| \omega^3 = 1 \left(x^2 + x + 1 = 0\right) 0 \left\langle \begin{matrix} \omega \\ \omega^2 \end{matrix} \right| \omega^3 = 1$ let $\alpha = \omega$

$(1+\alpha)^7 = 1 + {}^7c_1\alpha + {}^7c_2\alpha^2 + {}^7c_3\alpha^3 + {}^7c_4\alpha^4 + {}^7c_5\alpha^5 + {}^7c_6\alpha^6 + {}^7c_7\alpha^7$

$= 1 + 7\alpha + 21\alpha^2 + 35\alpha^3 + 35\alpha^4 + 24\alpha^5 + 7\alpha^6 + \alpha^9$

$\because \alpha = \omega$

$= 1 + 7\omega + 21\omega^2 + 35\omega^3 + 35\omega^4 + 21\omega^5 + 7\omega^6 + \omega^7$

$\left(\because \omega^3 = 1\right)$

$= 1 + 7\omega + 21\omega^2 + 35 + 35\omega + 21\omega + 7 + \omega$

$= 43 + 43\omega + 42\omega^2$

$= 43 + 43\alpha + 42\alpha^2 = A + B\alpha + C\alpha^2$

$\Rightarrow A = 43$ |B = 43| C = 42

5(129 – 86 – 42)

=5(129 – 128) = 5

35. Let the complex numbers α and $\frac{1}{\overline{\alpha}}$ lie on the circle $|z - z_0|^2 = 4$ and $|z - z_0|^2 = 16$ respectively, where $z_0 = 1 + i$. Then, the value of $100\,|\alpha|^2$ is _____. **[JEE(Main)Jan.2024]**

माना सम्मिश्र संख्याएँ α तथा $\frac{1}{\overline{\alpha}}$ क्रमशः वृत्तों $|z - z_0|^2 = 4$ तथा $|z - z_0|^2 = 16$ पर हैं, जहाँ $z_0 = 1 + i$ है। तो $100\,|\alpha|^2$ का मान है _____। **[JEE(Main)Jan.2024]**

Ans. 20

Sol. $|\alpha - z_0|^2 = 4$

$\Rightarrow (\alpha - z_0)(\overline{\alpha} - \overline{z}_0) = 4$

$|\alpha|^2 - \alpha\bar{z}_0 - \bar{\alpha}z_0 + |z_0|^2 = 4$

$|\alpha|^2 - \alpha\bar{z}_0 - \bar{\alpha}z_0 = 2$ _____(i)

(ii) $\left|\frac{1}{\bar{\alpha}} - z_0\right|^2 = 16 \Rightarrow (1-\bar{\alpha}z_0)(1-\alpha\bar{z}_0) = 16|\alpha|^2$

$\Rightarrow 1 - \alpha\bar{z}_0 - \bar{\alpha}z_0 + |\alpha|^2 \cdot 2 = 16|\alpha|^2$ _____(ii)

from (i) and (ii) $-1 - |\alpha|^2 = 2 - 16|\alpha|^2 \Rightarrow 15|\alpha|^2 = 3 \Rightarrow 5|\alpha|^2 = 1$

$100|\alpha|^2 = 20$

36. Let α, β the roots of the equation $x^2 - x + 2 = 0$ with $\text{Im}(\alpha) > \text{Im}(\beta)$. Then $\alpha^6 + \alpha^4 + \beta^4 - 5\alpha^2$ is equal to _____. **[JEE(Main)Jan.2024]**

माना समीकरण $x^2 - x + 2 = 0$ के मूल α, β हैं तथा $\text{Im}(\alpha) > \text{Im}(\beta)$ है। तो $\alpha^6 + \alpha^4 + \beta^4 - 5\alpha^2$ बराबर है_____. **[JEE(Main)Jan.2024]**

Ans. 13

Sol. $x^2 + 2 = x$

$\Rightarrow x^4 + 4 + 4x^2 = x^2$

So,

$x^4 = -4 - 3x^2$

$\Rightarrow x^6 = -4x^2 - 3x^4$

$\Rightarrow x^6 = 12 + 5x^2$

Now,

$\alpha^6 - 5\alpha^2 = 12$

So $\alpha^4 + \beta^4 + 12$

$= 4 - 3(\alpha^2 + \beta^2)$

$= 4 - 3\,[1 - 4]$

$= 13$

37. Let α, β be the roots of the equation $x^2 - \sqrt{6}x + 3 = 0$ such that $\text{Im}(\alpha) > \text{Im}(\beta)$. Let a, b be integers not divisible by 3 and n be a natural number such that $\frac{\alpha^{99}}{\beta} + \alpha^{98} = 3^n(a + ib)$, $i = \sqrt{-1}$. Then n + a + b is equal to **[JEE(Main)Jan.2024]**

माना समीकरण $x^2-\sqrt{6}x+3=0$ के मूल α, β हैं तथा Im (α) > Im (β) है। माना 3 से अभाज्य पूर्णांक a, b हैं तथा घन पूर्णांक n के लिए $\frac{\alpha^{99}}{\beta}+\alpha^{98}=3^n(a+ib)$, $i=\sqrt{-1}$ है। तो n + a + b बराबर है _____

[JEE(Main)Jan.2024]

Ans. 49

Sol. $x^2-\sqrt{6}x+3=0 \begin{matrix}\nearrow \alpha \\ \searrow \beta\end{matrix}$

$$x=\frac{\sqrt{6}\pm i\sqrt{6}}{2}=\frac{\sqrt{6}}{2}(1\pm i)$$

$$\alpha=\sqrt{3}\left(e^{i\frac{\pi}{4}}\right),\ \beta=\sqrt{3}\left(e^{-i\frac{\pi}{4}}\right)$$

$$\therefore \frac{\alpha^{99}}{\beta}+\alpha^{98}=\alpha^{98}\left(\frac{\alpha}{\beta}+1\right)$$

$$=\frac{\alpha^{98}(\alpha+\beta)}{\beta}=3^{49}\left(e^{i99\frac{\pi}{4}}\right)\times\sqrt{2}$$

$= 3^{49}(-1 + i)$
$= 3^n(a + ib)$
$\therefore n = 49, a = -1, b = 1$
$\therefore n + a + b = 49 - 1 + 1 = 49$

38. If α denotes the number of solutions of $|1-i|^x=2^x$ and $\beta=\left(\frac{|z|}{\arg(z)}\right)$, where $z=\frac{\pi}{4}(1+i)^4\left[\frac{1-\sqrt{\pi}i}{\sqrt{\pi}+i}+\frac{\sqrt{\pi}-i}{1+\sqrt{\pi}i}\right]$, $i=\sqrt{-1}$, then the distance of the point (α, β) from the line $4x-3y=7$ is _____.

[JEE(Main)Jan.2024]

यदि $|1-i|^x = 2^x$ के हलों की संख्या α है तथा $\beta=\left(\frac{|z|}{\arg(z)}\right)$ है, जहाँ $z=\frac{\pi}{4}(1+i)^4\left[\frac{1-\sqrt{\pi}i}{\sqrt{\pi}+i}+\frac{\sqrt{\pi}-i}{1+\sqrt{\pi}i}\right]$, $i=\sqrt{-1}$ है, तो js§kk $4x - 3y = 7$ से बिन्दु (α, β) की दूरी है _____.

[JEE(Main)Jan.2024]

Ans. 3

Sol. $$z=\frac{\pi}{4}(1+i)^4\left[\frac{1+\pi+\pi+1}{i(1+\pi)-\sqrt{\pi}+\sqrt{\pi}}\right]$$

$$=\frac{\pi}{4}(1+i)^4\frac{2}{i}=\frac{\pi}{2}\frac{(1+i)^4}{i}=\frac{-\pi}{2}i(1+i)^4$$

$$|z| = \left|\frac{-\pi}{2}\right||i||1+i|^4 = 2\pi$$

$$z = \frac{-\pi}{2}(-4i) = 2\pi i \Rightarrow \arg(z) = \frac{\pi}{2}$$

$$\Rightarrow \beta = \frac{2\pi}{\frac{\pi}{2}} = 4$$

Also $|1 - i|^x = 2^x$

$$\left(\sqrt{2}\right)^x = 2^x$$

$\Rightarrow x = 0$

$\Rightarrow$ 1 solution $\Rightarrow \alpha = 1$

Now perpendicular distance of $4x - 3y = 7$ from (1,4)

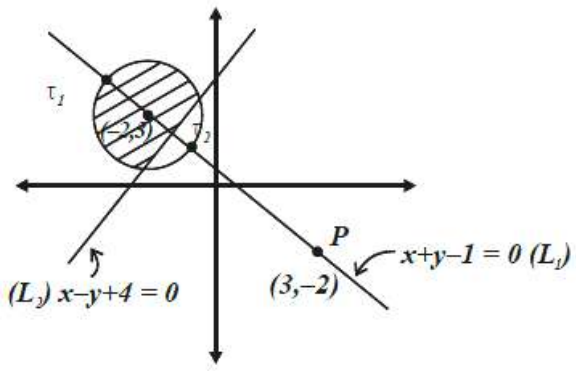
$$\left|\frac{4(1)-3(4)-7}{5}\right| = 3$$

39. Let $P = \{z \in C : |z + 2 - 3i| \le 1\}$ and $Q = \{z \in C : z(1+i) + \bar{z}(1-i) \le -8\}$. Let in $P \cap Q$, $|z - 3 + 2i|$ be maximum and minimum at z_1 and z_2 respectively. If $|z_1|^2 + 2|z_2|^2 = \alpha + \beta\sqrt{2}$, where α, β are integers, then $\alpha + \beta$ equals _____. **[JEE(Main)Jan.2024]**

माना $P = \{z \in C : |z + 2 - 3i| \le 1\}$ तथा $Q = \{z \in C : z(1+i) + \bar{z}(1-i) \le -8\}$ हैं। माना $P \cap Q$ में, $|z - 3 + 2i|$, z_1 तथा z_2 पर क्रमशः अधिकतम तथा निम्नतम है। यदि $|z_1|^2 + 2|z_2|^2 = \alpha + \beta\sqrt{2}$ है, जहाँ α, β पूर्णांक हैं, तो $\alpha + \beta$ बराबर है _____ **[JEE(Main)Jan.2024]**

Ans. 36

Sol.



Clearly for the shaded region z_1 is the intersection of the circle and the line passing through P (L_1) and z_2 is intersection of line L_1 & L_2

Circle : $(x + 2)^2 + (y - 3)^2 = 1$

$L_1 : x + y - 1 = 0$

$L_2 : x - y + 4 = 0$

On solving circle & L_1 we get

$z_1 : \left(-2-\frac{1}{\sqrt{2}}, 3+\frac{1}{\sqrt{2}}\right)$

On solving L_1 and z_2 is intersection of line L_1 & L_2

we get $z_2 : \left(\frac{-3}{2}, \frac{5}{2}\right)$

$|z_1|^2 + 2|z_2|^2 = 14 + 5\sqrt{2} + 17$

$= 31 + 5\sqrt{2}$

So, $\alpha = 31$

$\beta = 5$

$\alpha + \beta = 36$

40. Let α and β be the sum and the product of all the non-zero solutions of the equation $(\bar{z})^2 + |z| = 0$, $z \in C$. Then $4(\alpha^2 + \beta^2)$ is equal to : **[JEE(Main)April,2024]**

माना समीकरण $(\bar{z})^2 + |z| = 0$, $z \in C$ के सभी शून्येत्तर हलों के योग और गुणनफल क्रमशः α और β हैं। तो $4(\alpha^2 + \beta^2)$ बराबर है: **[JEE(Main)April,2024]**

(A) 2 (B) 8 (C) 4 (D) 6

Ans. C

Sol. $z = x + iy$

$\bar{z} = x - iy$

$\bar{z}^2 = x^2 - y^2 - 2ixy$

$\Rightarrow x^2 - y^2 - 2ixy + \sqrt{x^2 + y^2} = 0$

$\Rightarrow x = 0$ or $y = 0$

$-y^2 + |y| = 0$ $x^2 + |x| = 0$

$|y| = |y|^2$ $x = 0$

$y = 0, \pm 1$

$\Rightarrow i, -i$ $\Rightarrow \alpha = i - i = 0$

are roots $\beta = i(-i) = 1$

$4(0 + 1) = 4$

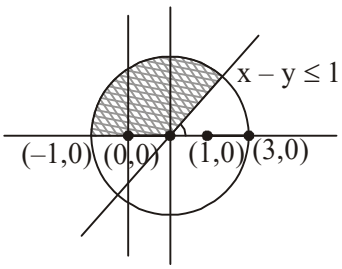
41. The area (in sq. units) of the region $S = \{z \in C : |z-1| \le 2; (z+\bar{z}) + i(z-\bar{z}) \le 2, \text{Im}(z) \ge 0\}$ is : **[JEE(Main)April,2024]**

क्षेत्र $S = \{z \in C : |z-1| \le 2; (z+\bar{z}) + i(z-\bar{z}) \le 2, \text{Im}(z) \ge 0\}$ का क्षेत्रफल (वर्ग इकाईयों में) है :

(A) $\frac{7\pi}{4}$ (B) $\frac{7\pi}{3}$ (C) $\frac{17\pi}{8}$ (D) $\frac{3\pi}{2}$

Ans. D

Sol.



$|z-1| \le 2 \Rightarrow (x-1)^2 + y^2 = 4$

$(z+\bar{z})+i(z-\bar{z}) \le 2 \Rightarrow x - y \le 1$

$\text{Im}\,|z| \ge 0 \Rightarrow y \ge 0$

$$\text{Area of region} = \frac{1}{2}(2)^2\left(\frac{3\pi}{4}\right) = \frac{3\pi}{2}$$

42. Consider the following two statements

Statement I : For any two non-zero complex numbers z_1, z_2.

$$(|z_1|+|z_2|)\left|\frac{z_1}{|z_1|}+\frac{z_2}{|z_2|}\right| \le 2(|z_1|+|z_2|) \text{ and}$$

Statement II : If x, y, z are three distinct complex numbers and a, b, c are three positive real numbers such that $\frac{a}{|y-z|} = \frac{b}{|z-x|} = \frac{c}{|x-y|}$, then $\frac{a^2}{y-z}+\frac{b^2}{z-x}+\frac{c^2}{x-y} = 1$.

Between the above two statemetns : **[JEE(Main)April,2024]**

(A) Statement I is correct but Statement II is incorrect

(B) Statement I is incorrect but Statement II is correct

(C) Both Statement I and Statement II are incorrect

(D) Both Statement I and Statement II are correct

कथनों

कथन I : किसी भी दो शून्येत्तर सम्मिश्र संख्याओं z_1, z_2 के लिए

$$(|z_1|+|z_2|)\left|\frac{z_1}{|z_1|}+\frac{z_2}{|z_2|}\right| \le 2(|z_1|+|z_2|)$$ तथा

कथन II : यदि x, y, z तीन भिन्न सम्मिश्र संख्याएँ हैं और a, b, c तीन धनात्मक वास्तविक संख्याएँ हैं ताकि $\frac{a}{|y-z|}=\frac{b}{|z-x|}=\frac{c}{|x-y|}$

है, तो $\frac{a^2}{y-z}+\frac{b^2}{z-x}+\frac{c^2}{x-y}=1$ है।

[JEE(Main)April,2024]

(A) कथन I सही है, परन्तु कथन II गलत है

(B) कथन I गलत है परन्तु कथन II सही हैं

(C) कथन I तथा कथन II दोनों गलत है

(D) कथन I तथा कथन II दोनों सही है

Ans. A

Sol. **(I)** $(|Z_1|+|Z_2|)\left|\frac{Z_1}{|Z_1|}+\frac{Z_2}{|Z_2|}\right|$

$$\left|\frac{Z_1}{|Z_1|}+\frac{Z_2}{|Z_2|}\right| \le \left|\frac{Z_1}{|Z_1|}\right|+\left|\frac{Z_2}{|Z_2|}\right| = 1+1 = 2$$

(II) $\frac{a}{|y-z|}=\frac{b}{|z-x|}=\frac{c}{|x-y|}$

$$\frac{a^2}{(y-z)}\times\left(\frac{\overline{y-z}}{y-z}\right)+\frac{b^2}{z-x}\times\frac{\overline{z-x}}{z-x}+\frac{c^2}{x-y}\times\frac{\overline{x-y}}{x-y}$$

$$=\frac{a^2}{|y-z|^2}(\bar{y}-\bar{z}+\bar{z}-\bar{x}+\bar{x}-\bar{y})=0$$

So not true

43. Let $S_1=\{z\in C:|z|\le 5\}, S_2\left\{z\in C:\text{Im}\left(\frac{z+1-\sqrt{3}i}{1-\sqrt{3}i}\right)\ge 0\right\}$ and $S_3=\{z\in C:\text{Re}(z)\ge 0\}$. Then the area of the region $S_1\cap S_2\cap S_3$ is :

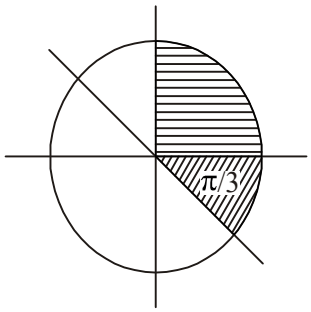
[JEE(Main)April,2024]

माना $S_1 = \{z \in C : |z| \le 5\}, S_2 \left\{z \in C : \text{Im}\left(\frac{z+1-\sqrt{3}i}{1-\sqrt{3}i}\right) \ge 0\right\}$ तथा $S_3 = \{z \in C : \text{Re}(z) \ge 0\}$ हैं। तो क्षेत्र $S_1 \cap S_2 \cap S_3$ का क्षेत्रफल है : **[JEE(Main)April,2024]**

(A) $\frac{125\pi}{24}$ (B) $\frac{125\pi}{4}$ (C) $\frac{125\pi}{12}$ (D) $\frac{125\pi}{6}$

Ans. C

Sol.



$S_1 : x^2 + y^2 \le 5^2 ; S_2 : \sqrt{3}\ x + y \ge 0 ; S_3 : x \ge 0$

$$\frac{z+1-i\sqrt{3}}{1-\sqrt{3}i} = \frac{\left((x+1)+i\left(y-\sqrt{3}\right)\left(1+i\sqrt{3}\right)\right)}{1+3}$$

$$I_m(z) = (x+1)\sqrt{3} + \left(y-\sqrt{3}\right) = \sqrt{3}x + y$$

$$S_1 \cap S_2 \cap S_3 = \frac{1}{4}\pi 5^2 + \frac{1}{2}5^2\frac{\pi}{3} = \frac{125\pi}{12}$$

44. If z_1, z_2 are two distinct complex number such that $\left|\frac{z_1 - 2z_2}{\frac{1}{2} - z_1\bar{z}_2}\right| = 2$, then : **[JEE(Main)April,2024]**

(A) both z_1 and z_2 lie on the same circle

(B) z_1 lies on a circle of radius $\frac{1}{2}$ and z_2 lies on a circle of radius 1

(C) either z_1 lies on a circle of radius 1 or z_2 lies on a circle of radius $\frac{1}{2}$

(D) either z_1 lies on a circle of radius $\frac{1}{2}$ or z_2 lies on a circle of radius 1

यदि z_1, z_2 दो भिन्न सम्मिश्र संख्याऐं इस प्रकार है कि $\left|\frac{z_1-2z_2}{\frac{1}{2}-z_1\bar{z}_2}\right|=2$ है, तो : **[JEE(Main)April,2024]**

(A) z_1 तथा z_2 दोनों समान वृत्त पर स्थित हैं

(B) z_1, $\frac{1}{2}$ त्रिज्या के वृत्त पर तथा z_2, 1 त्रिज्या के वृत्त पर स्थित है

(C) या तो z_1, 1 त्रिज्या के वृत्त पर या z_2, $\frac{1}{2}$ त्रिज्या के वृत्त पर स्थित है

(D) या तो z_1, $\frac{1}{2}$ त्रिज्या के वृत्त पर या z_2, 1 त्रिज्या के वृत्त पर स्थित है

Ans. C

Sol. $\left|\frac{z_1-2z_2}{\frac{1}{2}-z_1\bar{z}_2}\right|=2$

$\Rightarrow |z_1-2z_2|=|1-2z_1\bar{z}_2|$

$\Rightarrow |z_1-2z_2|^2|1-2z_1\bar{z}_2|^2$

$\Rightarrow |z_2|^2-2z_1\bar{z}_2-2\bar{z}_1z_2+4|z_2|^2=1-2\bar{z}_1z_2-2z_1\bar{z}_2+4|z_1|^2|z_2|^2$

$\Rightarrow |z_1|^2-1+4|z_2|^2-4|z_1|^2|z_2|^2=0$

$\Rightarrow \left(|z_1|^2-1\right)\left(1-4|z_2|^2\right)=0$

either $|z_1|^2-1=0$ or $1-4|z_2|^2=0$

$|z_1|=1$ or $|z_2|=\frac{1}{2}$

$\therefore$ either z_1 lies on circle of radius 1

or z_2 lies on circle of radius $\frac{1}{2}$

45. Let z be a complex number such that $|z+2|=1$ and $\text{Im}\left(\frac{z+1}{z+2}\right)=\frac{1}{5}$. Then the value of $\left|\text{Re}\left(\overline{z+2}\right)\right|$ is :

[JEE(Main)April,2024]

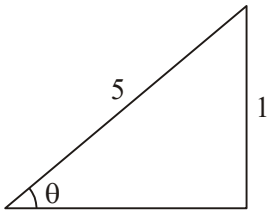
माना z एक समिश्र संख्या इस प्रकार है कि $|z+2|=1$ और $\text{Im}\left(\frac{z+1}{z+2}\right)=\frac{1}{5}$ तब $\left|\text{Re}\left(\overline{z+2}\right)\right|$ का मान होगा :

(A) $\frac{2\sqrt{6}}{5}$ (B) $\frac{24}{5}$ (C) $\frac{1+\sqrt{6}}{5}$ (D) $\frac{\sqrt{6}}{5}$

Ans. A

Sol. Let $z+2=e^{i\theta} \Rightarrow \left(\frac{z+1}{z+2}\right)=\frac{-2+e^{i\theta}+1}{e^{i\theta}}=1-\frac{1}{e^{i\theta}}$

$\Rightarrow \left(\frac{z+1}{z+2}\right)=1-\cos\theta+i\sin\theta$



$\sin\theta=\frac{1}{5}$

$\text{Re}(z+2)=\cos\theta \Rightarrow \left|\text{Re}\left(\overline{z+2}\right)\right|=\frac{2\sqrt{6}}{5}$

46. If the set $R=\{(a, b) : a+5b=42, a, b \in N\}$ has m elements and $\sum_{n=1}^{m}\left(1-i^{n!}\right)=x+iy$, where $i=\sqrt{-1}$, then the value of $m+x+y$ is : **[JEE(Main)April,2024]**

यदि समुच्चय $R = \{(a, b) : a + 5b = 42, a, b \in N\}$ में m अवयव है और $\sum_{n=1}^{m}\left(1-i^{n!}\right)=x+iy$, जहाँ $i=\sqrt{-1}$, तब $m + x + y$ का मान होगा : **[JEE(Main)April,2024]**

(A) 8 (B) 12 (C) 5 (D) 4

Ans. B

Sol. $a + 5b = 42, a, b \in N$

$a = 42 – 5b, b = 1, a = 37$

$b = 2, a = 32$

$b = 3, a = 27$

⋮

$b = 8, a = 2$

R has "8" elements $\Rightarrow m=8$

$\sum_{n=1}^{8}\left(1-i^{n!}\right)=x+iy$

for $n \geq 4$, $i^{n!}=1$

$\Rightarrow (1-i)+(1-i^{2!})+(1-i^{3!})$

$=1-i+2+1+1$

$=5-i=x+iy$

$m+x+y=8+5-1=12$

47. The sum of all possible values of $\theta \in [-\pi, 2\pi]$, for which $\frac{1+i\cos\theta}{1-2i\cos\theta}$ is purely imaginary, is equal to :

[JEE(Main)April,2024]

$\theta \in [-\pi, 2\pi]$ के सभी संभव मानों, जिसके लिए $\frac{1+i\cos\theta}{1-2i\cos\theta}$ शुद्धतः काल्पनिक है, का योग बराबर है:

[JEE(Main)April,2024]

(A) 2π (B) 5π (C) 3π (D) 4π

Ans. C

Sol. $z=\frac{1+i\cos\theta}{1-2i\cos\theta}$

$z=\bar{z} \Rightarrow \frac{1+i\cos\theta}{1-2i\cos\theta}=-\left(\frac{\overline{1+i\cos\theta}}{\overline{1-2i\cos\theta}}\right)$

$(1+i\cos\theta)\left(\overline{1-2i\cos\theta}\right)=-(1-2i\cos\theta)\left(\overline{1+i\cos\theta}\right)$

$\Rightarrow 1+3i\cos\theta-2\cos^2\theta=-(1-3i\cos\theta-2\cos^2\theta)$

$\Rightarrow \cos^2\theta=\frac{1}{2} \Rightarrow \theta=\pm\frac{\pi}{4},\pm\frac{3\pi}{4},\frac{5\pi}{4},\frac{7\pi}{4}$

Sum $\Rightarrow 3\pi$

48. Let z be a complex number such that the real part of $\frac{z-2i}{z+2i}$ is zero. Then, the maximum value of $|z-(6+8i)|$ is equal to :

[JEE(Main)April,2024]

माना z एक सम्मिश्र संख्या है, जिसके लिए $\frac{z-2i}{z+2i}$ का वास्तविक भाग शून्य है। तो $|z-(6+8i)|$ का अधिकतम मान बराबर है :

[JEE(Main)April,2024]

(A) ∞ (B) 8 (C) 10 (D) 12

Ans. D

Sol. $\frac{z-2i}{z+2i}+\frac{\bar{z}+2i}{\bar{z}-2i}=0$

$z\bar{z} - 2i\bar{z} - 2iz + 4(-1) + z\bar{z} + 2zi + 2\bar{z}i + 4(-1) = 0$

$\Rightarrow 2|z|^2 = 8$

$\Rightarrow |z| = 2$

$|z - (6 + 8i)|_{\text{maximum}} = 10 + 2 = 12$

49. Let x_1, x_2, x_3, x_4 be the solution of the equation $4x^4 + 8x^3 - 17x^2 - 12x + 9 = 0$ and $\left(4 + x_1^2\right)\left(4 + x_2^2\right)\left(4 + x_3^2\right)\left(4 + x_4^2\right) = \frac{125}{16}m$. Then the value of m is _____.

[JEE(Main)April,2024]

माना x_1, x_2, x_3, x_4 समीकरण $4x^4 + 8x^3 - 17x^2 - 12x + 9 = 0$ के हल हैं तथा $\left(4 + x_1^2\right)\left(4 + x_2^2\right)\left(4 + x_3^2\right)\left(4 + x_4^2\right) = \frac{125}{16}m$ है। तो m का मान है _____ **[JEE(Main)April,2024]**

Ans. 221

Sol. $4x^4 + 8x^3 - 17x^2 - 12x + 9 = 0$

x_1, x_2, x_3, x_4 are the solution of equation

$4x^4 + 8x^3 - 17x^2 - 12x + 9 = 4(x - x_1)(x - x_2)(x - x_3)(x - x_4)$

put $x = 2i$

$4(2i)^4 + 8(2i)^3 - 17(2i)^2 - 12(2i) + 9$

$= 4[(2i - x_1)(2i - x_2)(2i - x_3)(2i - x_4)]$

$\Rightarrow$ $64 - 64i + 68 - 24i + 9$

$\Rightarrow$ $|141 - 88i| = 4|(2i - x_1)(2i - x_2)(2i - x_3)(2i - x_4)|$

$\Rightarrow$ $(141)^2 + (88)^2 = 16[(4 + x_1^2)(4 + x_2^2)(4 + x_3^2)(4 + x_4^2)]$

$\Rightarrow$ $\frac{19881 + 7744}{16} = \left(4 + x_1^2\right)\left(4 + x_2^2\right)\left(4 + x_4^2\right)$

$\Rightarrow$ $\frac{27625}{16} = \frac{125}{16}m$

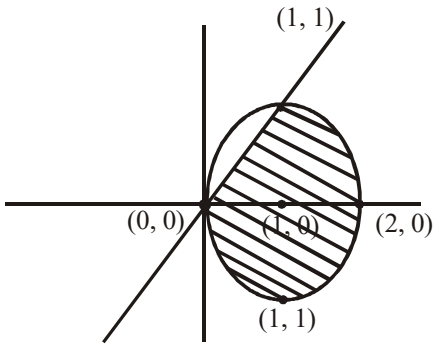
$m = 221$

50. The sum of the square of the modulus of the elements in the set $\{z = a + ib : a, b \in Z, z \in C, |z - 1| \le 1, |z - 5| \le |z - 5i|\}$ is _____. **[JEE(Main)April,2024]**

समुच्चय $\{z = a + ib : a, b \in Z, z \in C, |z - 1| \le 1, |z - 5| \le |z - 5i|\}$ के अवयवों के मापांक के वर्गों का योग है _____.

Ans. 9

Sol.



$|Z-1| \le 1$

$|Z-5| \le |Z-5i|$

$(a, b) = \{(0, 0), (1, 1), (1, 0), (1, -1), (2, 0)\}$

$|Z|^2 = \{0, 2, 1, 2, 4|$

Sum = 9